Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Count the Number of Words in a Text File
Write a program to read a text file and count the total number of words present in it.
Q2. Display Student Marks Using a Dictionary
Write a program to store student names and their marks in a dictionary and display the name and marks of each student.
Q3. Find the Highest Mark from a CSV File
Write a program to read student names and their marks from week_6_set1_q3.csv and display the name of the student who scored the highest mark.
Q4. Count the Frequency of Each Word in a Text File
Write a program to read words from a text file and use a dictionary to count how many times each word appears.
Q5. Display Products with Low Stock from a CSV File
Write a program to read product names and their stock quantities from week_6_set1_q5.csv and display only the products whose stock is less than 10.
Q6. Update Student Marks in a Dictionary
Write a program to store student names and marks in a dictionary, update the marks of a given student, and display the updated dictionary.
Q7. Copy Lines Containing a Given Word into Another File
Write a program to read a text file and copy only the lines containing a user-entered word into another text file.
Q8. Store Names and Ages in a CSV File
Write a Python program that asks the user to enter the names and ages of five people. Store the data in a list of lists and write the results into a CSV file named week_6_set1_q8.csv.
Solutions
1. Count the Number of Words in a Text File

```
file = open("story.txt", "w")
file.write("Python is easy to learn")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
text = file.read()
file.close()
```

```
words = text.split()
count = len(words)

print("Total number of words:", count)
```

Output:

```
Enter file name: story.txt
Total number of words: 5
```

2. Display Student Marks Using a Dictionary

```
students = {}

n = int(input("Enter number of students: "))

for i in range(n):
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    students[name] = marks

print("Student Marks:")

for name in students:
    print(name, ":", students[name])
```

Output:

```
Enter number of students: 3
Enter student name: Aarav
Enter marks: 78
Enter student name: Diya
Enter marks: 92
Enter student name: Kabir
Enter marks: 85
Student Marks:
Aarav : 78
Diya : 92
Kabir : 85
```

3. Find the Highest Mark from a CSV File

```
import csv

file = open("week_6_set1_q3.csv", "r")
reader = csv.DictReader(file)

highest_name = ""
highest_mark = -1
```

```
for row in reader:
    name = row["Student Name"]
    marks = int(row["Marks"])

    if marks >highest_mark:
highest_mark = marks
highest_name = name

file.close()

print("Student with highest mark:", highest_name)
print("Highest mark:", highest_mark)
```

Output:

```
Student with highest mark: Diya
Highest mark: 92
```

4. Count the Frequency of Each Word in a Text File

```
file = open("words.txt", "w")
file.write("python java python c java python")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
text = file.read()
file.close()

words = text.split()
frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

print("Word Frequency:")

for word in frequency:
    print(word, ":", frequency[word])
```

Output:

```
Enter file name: words.txt
```

Word Frequency:
python : 3
java : 2
c : 1

5. Display Products with Low Stock from a CSV File

```
import csv

file = open("week_6_set1_q5.csv", "r")
reader = csv.DictReader(file)

print("Products with stock less than 10:")

for row in reader:
    name = row["Product Name"]
    stock = int(row["Stock Quantity"])

    if stock < 10:
        print(name, ":", stock)

file.close()
```

Output:

Products with stock less than 10:
Pen : 7
Backpack : 4
Eraser : 9

6. Update Student Marks in a Dictionary

```
students = {
    "Aarav": 78,
    "Diya": 92,
    "Kabir": 85
}

print("Original dictionary:", students)

name = input("Enter student name to update: ")
new_mark = int(input("Enter new mark: "))

if name in students:
    students[name] = new_mark
    print("Updated dictionary:", students)
else:
    print("Student not found")
```

Output:

```
Original dictionary: {'Aarav': 78, 'Diya': 92, 'Kabir': 85}
Enter student name to update: Aarav
Enter new mark: 95
Updated dictionary: {'Aarav': 95, 'Diya': 92, 'Kabir': 85}
```

7. Copy Lines Containing a Given Word into Another File

```
file = open("input.txt", "w")
file.write("Python is easy to learn\n")
file.write("Java is a programming language\n")
file.write("Python is used for data analysis\n")
file.write("C is a programming language\n")
file.close()

file_name = input("Enter file name: ")
search_word = input("Enter word to search: ")

source = open(file_name, "r")
destination = open("filtered.txt", "w")

for line in source:
    if search_word.lower() in line.lower():
destination.write(line)

source.close()
destination.close()

destination = open("filtered.txt", "r")
print("Matching lines:")
print(destination.read(), end="")
destination.close()
```

Output:

```
Enter file name: input.txt
Enter word to search: Python
Matching lines:
Python is easy to learn
Python is used for data analysis
```

8. Store Names and Ages in a CSV File

```
import csv

people = []

for i in range(5):
```

```
    name = input("Enter name: ")
    age = int(input("Enter age: "))
people.append([name, age])

file = open("week_6_set1_q8.csv", "w", newline="")
writer = csv.writer(file)

writer.writerow(["Name", "Age"])
writer.writerows(people)

file.close()

print("Data written to week_6_set1_q8.csv")

file = open("week_6_set1_q8.csv", "r")
reader = csv.reader(file)


for row in reader:
    print(row[0], row[1])

file.close()
```

Output:

```
Enter name: Aarav
Enter age: 22
Enter name: Diya
Enter age: 19
Enter name: Kabir
Enter age: 25
Enter name: Meera
Enter age: 21
Enter name: Rohan
Enter age: 23
Data written to week_6_set1_q8.csv
Name, Age
Name Age
Aarav 22
Diya 19
Kabir 25
Meera 21
Rohan 23
```

Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Prepared By,
Prof. Adithi